

Probability & Probability Distributions

Formative Assessment 4

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Section: APM1110 - SECTION 2

GitHub Link:

Exercise 7.1

Item 2

A binary communication channel carries data as one of two sets of signals denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1, and a transmitted 1 is sometimes received as a 0. For a given channel, it can be assumed that a transmitted 0 is correctly received with probability 0.95, and a transmitted 1 is correctly received with probability 0.75. Also, 70% of all messages are transmitted as a 0.

(a) Probability that a 1 was received

Given:

- $P(T = 0) = 0.70$
- $P(T = 1) = 0.30$
- $P(R = 1 \mid T = 0) = 0.05$
- $P(R = 1 \mid T = 1) = 0.75$

Using the Law of Total Probability:

$$P(R = 1) = P(R = 1|T = 0)P(T = 0) + P(R = 1|T = 1)P(T = 1)$$

```
P_T0 <- 0.70
P_T1 <- 0.30

P_R1_given_T0 <- 1 - 0.95
P_R1_given_T1 <- 0.75

P_R1 <- P_R1_given_T0 * P_T0 +
        P_R1_given_T1 * P_T1

P_R1
```

```
## [1] 0.26
```

Therefore, the probability that a 1 was received is:

$$P(R = 1) = 0.26 = 26\%$$

(b) Probability that a 1 was transmitted given that a 1 was received

Using Bayes' Theorem:

$$P(T = 1|R = 1) = \frac{P(R = 1|T = 1)P(T = 1)}{P(R = 1)}$$

```
P_T1_given_R1 <- (P_R1_given_T1 * P_T1) / P_R1
```

```
P_T1_given_R1
```

```
## [1] 0.8653846
```

Therefore:

$$P(T = 1|R = 1) \approx 0.8654 = 86.54\%$$

Item 7

There are three employees working at an IT company: Jane, Amy, and Ava, doing 10%, 30%, and 60% of the programming, respectively. 8% of Jane's work, 5% of Amy's work, and just 1% of Ava's work is in error.

Overall percentage of error

Given:

- $P(\text{Jane}) = 0.10$
- $P(\text{Amy}) = 0.30$
- $P(\text{Ava}) = 0.60$
- $P(\text{Error} | \text{Jane}) = 0.08$
- $P(\text{Error} | \text{Amy}) = 0.05$
- $P(\text{Error} | \text{Ava}) = 0.01$

Using the Law of Total Probability:

$$P(\text{Error}) = P(\text{Error}|\text{Jane})P(\text{Jane}) + P(\text{Error}|\text{Amy})P(\text{Amy}) + P(\text{Error}|\text{Ava})P(\text{Ava})$$

```

P_Jane <- 0.10
P_Amy <- 0.30
P_Ava <- 0.60

P_Error_Jane <- 0.08
P_Error_Amy <- 0.05
P_Error_Ava <- 0.01

P_Error <- P_Error_Jane * P_Jane +
           P_Error_Amy * P_Amy +
           P_Error_Ava * P_Ava

P_Error

```

```
## [1] 0.029
```

The overall percentage of error is:

```
P_Error * 100
```

```
## [1] 2.9
```

Therefore:

$$P(Error) = 0.029 = 2.9\%$$

Most likely person to have written the program

Using Bayes' Theorem:

```

P_Jane_given_Error <- (P_Error_Jane * P_Jane) / P_Error
P_Amy_given_Error <- (P_Error_Amy * P_Amy) / P_Error
P_Ava_given_Error <- (P_Error_Ava * P_Ava) / P_Error

P_Jane_given_Error

```

```
## [1] 0.2758621
```

```
P_Amy_given_Error
```

```
## [1] 0.5172414
```

```
P_Ava_given_Error
```

```
## [1] 0.2068966
```

The probabilities are:

```
P_Jane_given_Error * 100
```

```
## [1] 27.58621
```

```
P_Amy_given_Error * 100
```

```
## [1] 51.72414
```

```
P_Ava_given_Error * 100
```

```
## [1] 20.68966
```

Therefore:

- Jane = 27.59%
- Amy = 51.72%
- Ava = 20.69%

Since Amy has the highest probability, **Amy is the most likely person to have written the program with an error.**

Amy, 51.72%
